

Demonstration of a Collision Control Mechanism for Inter-AIDC Traffic in a Spine-Leaf Multi-Granularity All-Optical Switching Network

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Abstract: We present a collision control mechanism for a spine-leaf multi-granularity all-optical switching network handling time-sensitive inter-AIDC traffic. An FPGA-based testbed demonstrate that the proposed mechanism effectively resolves optical time-slot conflicts and enhances overall network throughput. © 2025 The Author(s)

1. Introduction

The growing scale of artificial intelligence (AI) model training necessitates tight synchronization across distributed GPU clusters to ensure computational consistency. This synchronization enforces frequent communication among data centers (DCs), which generate deterministic yet bursty inter-AIDC traffic that coexists with conventional background traffic [1]. Such workloads require high-throughput and low-latency network to accelerate the training speed. Electrical packet switching based on the store-and forward mechanism is widely adopted in today's datacenter networks. It enables per-packet processing but suffers from latency and energy efficiency, limiting its performance and scalability. Motivated by these challenges, we proposed a spine-leaf multi-granularity all-optical switching network [2] that seamlessly combines wavelength and time-slot-level optical switching. As illustrated in Fig. 1, the network architecture consists of access layers that aggregate AIDC communication traffic through gateway nodes (GWs) and a core all-optical switching layers that interconnects multiple AIDCs. To overcome the lack of optical buffer, the system adopts a separation of data and control channels in the GWs. The control channel incorporates CDR and timestamp transmission [3, 4] to achieve global time and frequency synchronization. It enables precise optical time-slot routing and scheduling across the network. Meanwhile, by decoupling control from data, the data channel achieves flexible scheduling for contention-free optical interconnection among AIDCs. It performs framing, encapsulation, and aggregation of hierarchical traffic, which is subsequently organized and scheduled based on its characteristic. Building on this foundation, we experimentally validated a hybrid central and distributed control plane, as well as some resource allocation mechanisms [3, 5]. Furthermore, a centralized collision control mechanism [6] was proposed to perform global routing and scheduling with contention free. It achieved efficient coordination under stable traffic, but relies on a global network state collection and suffers from inefficient bandwidth utilization, making it less adaptable to bursty traffic. To improve high scalability and fast responsiveness, a distributed collision control mechanism that coordinates routing and resource allocation across multiple domains is required.

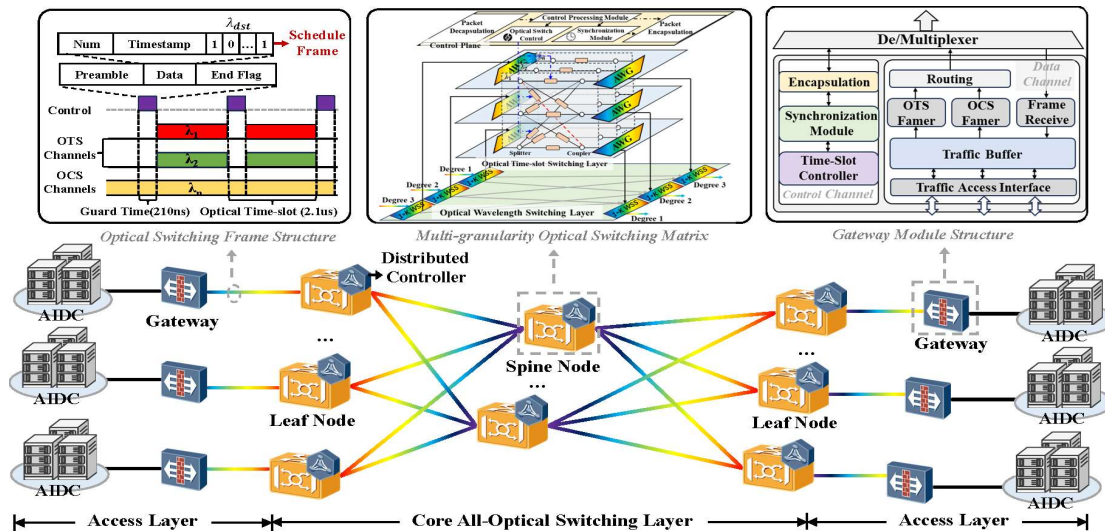


Fig. 1. A spine-leaf multi-granularity all-optical switching network architecture for inter-AIDC connection.

In this paper, we propose a distributed collision control mechanism for spine-leaf multi-granularity all-optical switching networks. It employs distributed controllers to make resilient routing, classifies inter-AIDC traffic as time-sensitive (TS) and conventional background traffic as best-effort (BE), and dynamically reroutes BE traffic based on the network status when collisions occur. Experimental results verify that the proposed mechanism effectively resolves optical time-slot contention, improving TS latency by 23.91% and overall network throughput by 12.60%.

2. Distributed Collision Control Mechanism in a Spine-leaf All-optical Switching Network

The communication phases of multiple training tasks in inter-AIDCs often overlap, resulting in intensified traffic bursts. These time-dependent communication phases generate consecutive surges of TS traffic. When such traffic share transmission paths with BE traffic and converge at the same switching node toward the destination, contention occurs, as illustrated in Fig. 2(a). The proposed collision control mechanism utilizes the rapid response of distributed controller at each switching node to detect traffic contention and dynamically select relay routes for BE traffic and reassigns it to an alternative path among multiple candidates generated in the spine-leaf topology based on network state. The selected BE traffic is forwarded to a relay node (e.g., GW2) through an appropriate output port on the spine switch. Upon reception, relay node monitors its local load and allocates a resilient forwarding path for retransmission, thereby mitigating congestion and improving network utilization. Since any GW may serve as a relay node, each must implement stringent queue management and scheduling control to ensure stability. The traffic processing of uplink and downlink at GWs is illustrated in Fig. 2(b). ①**Traffic Access**: TS traffic is received at GW2, where a transmission path is assigned and encapsulated into a fixed-length switching frame along with its corresponding control packet before entering the all-optical switching network. ②**Resilient Process**: The distributed controller at Spine 1 detects a convergence conflict between BE traffic from GW1 and TS traffic from GW2 in time-slot m of the wavelength λ_1 . Based on the current network state, Spine 1 designates GW2 as a relay node to receive the BE traffic, which is temporarily buffered according to its routing information. To prevent packet reordering, the control layer of the Spine 1 consistently routes subsequent packets through the same relay node. ③**Resilient Routing**: After classification, the relayed BE traffic enters the relay buffer. A weighted-based queue scheduling strategy is employed. The service weight (W) of each queue, calculated in real time (Eq. (1)), is used to balance load and latency between BE and TS traffic and to allocate resilient routing for subsequent retransmission.

$$W = \alpha \cdot \frac{R_{dmin}}{R_d} + \beta \cdot \frac{C_F}{C_{max}} + \gamma \cdot \frac{T}{T_{max}} \quad (1)$$

W comprises three normalized components, each weighted by a coefficient (α, β, γ) satisfying $\alpha + \beta + \gamma = 1$. The ratio R_{dmin}/R_d reflects the influence of traffic latency requirement, where R_{dmin} denotes the strictest latency requirement in the network, and R_d corresponds to the current queue. The ratio C_F/C_{max} represents buffer occupancy, with C_F denoting the queue length and C_{max} indicates the maximum buffer capacity. To mitigate starvation of low-priority BE traffic, T/T_{max} is introduced, where T indicates the elapsed waiting time since the last transmission and T_{max} denotes the maximum tolerable delay. ④**Receive Traffic**: The GW2 receives traffic destined for the server, reconstructs the original frame structure, and transmits it through fixed output ports.

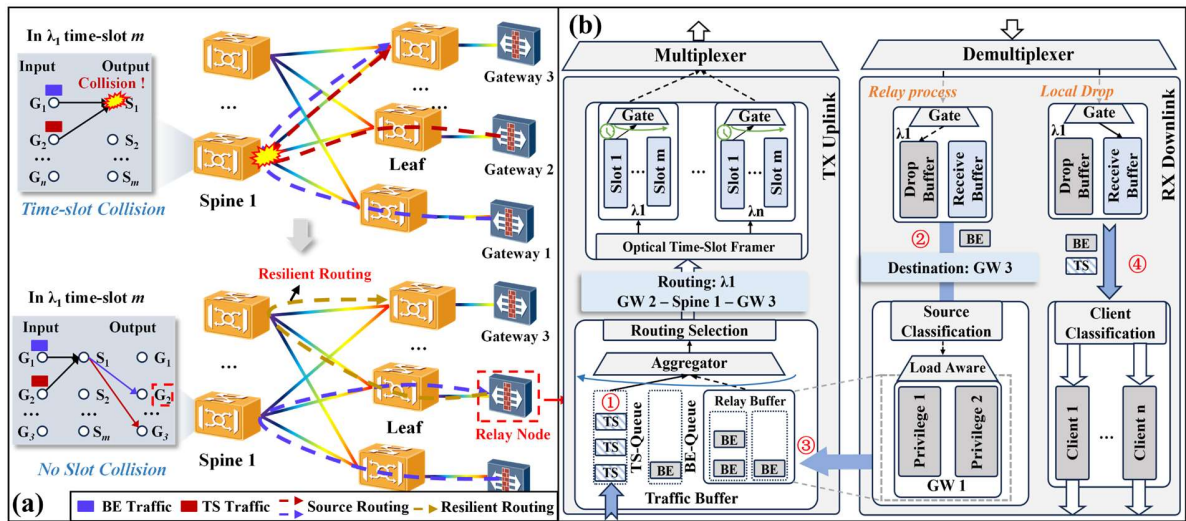


Fig. 2. (a) An illustration of distributed collision control mechanism; (b) The queue scheduling at the relay node.

3. Experimental Setup and Result Analysis

Figure 3(a) shows a simple topology of the FPGA-based testbed for verification, while Fig. 3(b) shows its implementation. We use semiconductor optical amplifiers (SOA) as the core components of the fast optical switch matrix. We set each 10 Gbps transceiver connects to the SOA switching matrix via optical fiber, and the capacity of each queue in GW is set to 64KB. The time-slot length is set $2.1\mu\text{s}$ with a guard time of 210ns [6]. The Ethernet Test Center generates two traffic types: the bursty TS traffic and continuous BE traffic. The TS traffic emulates communication traffic of inter-AIDC with an average rate of 500 Mbps [1]. BE traffic is divided into BE1 and BE2, with a load ratio of 2:1. To validate performance gains, the proposed mechanism is compared with the longest-queue-first scheme, which is a representative distributed scheduling strategy. A normalized load excluding packet overhead and guard time is adopted for performance analysis. Fig. 3(c-d) respectively show the average latency of TS traffic and its CDF when total traffic load is 0.6. The latency performance of TS traffic depends solely on the parameter α . As α increases, the average latency of TS traffic decreases and the upper bound of its CDF become smaller. Fig. 3(e-f) respectively show the average latency of BE1 and BE2 traffic. Since BE traffic must be buffered and retransmitted when colliding with TS traffic, its average latency is higher than that of the baseline. The latency performance of BE traffic is negatively correlated with α and positively correlated with β . When β and γ are smaller than α , BE traffic experiences longer buffering time, which ultimately leads to a sharp increase in the average BE latency when the load exceeds 0.5. Owing to its lower scheduling priority (calculated according to Eq. (1)), BE2 incurs a higher average latency than BE1. Fig. 3(g-h) respectively show the BE traffic packet loss rate and network throughput. When the load is low (e.g., 0.1), the queues can accommodate all BE traffic without overflow, leading to a very low packet loss rate. As the load increases (greater than 0.2), the proportion of BE traffic relative to TS traffic grows, and a large number of BE packets are buffered. This causes transient bursts where the instantaneous traffic exceeds the storage capacity, leading to a sharp increase in the packet loss rate at a load of 0.2 compared with that at 0.1. The packet loss rate of BE traffic affects the overall network throughput. The latency of TS traffic remains relatively stable across parameter variations, while BE traffic is more sensitive, exhibiting higher latency and packet loss. The optimal configuration is $\alpha = 0.33$, $\beta = 0.50$, and $\gamma = 0.17$. Under this setting, when BE packet loss stays below 0.01 at a normalized load of 0.6, the proposed mechanism improves TS latency by 23.91%, with latency increase of 5.53% and 10.92% for BE1 and BE2, respectively. Overall network throughput is enhanced by 12.60%.

4. Conclusion: We proposed a collision control mechanism for spine-leaf multi-granularity all-optical switching network. Experimental results showed a 23.91% latency reduction for time-sensitive inter-AIDC traffic and a 12.60% improvement for overall throughput, with only a minor increased in BE traffic latency.

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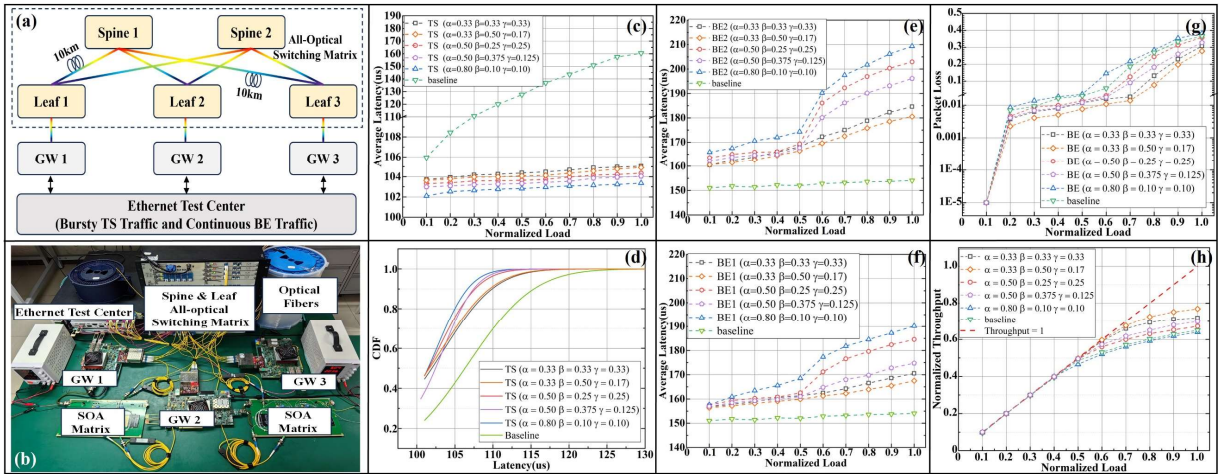


Fig. 3. (a) Overview of network setup; (b) Experimental testbed; (c) Average Latency of TS traffic; (d) CDF with TS traffic when load equals 0.6; (e) Average Latency of BE1 traffic; (f) Average Latency of BE2 traffic; (g) Packet loss rate with BE traffic; (h) Network throughput.